



SOMAIYA
VIKRAMJIJI UNIVERSITY

28 NOV 2023 (M)

Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: L.Y.	Semester: VII (SVU 2020)
Programme: B. Tech. (Computer Engineering)		
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: 116U01C701	Name of the Course: Software Architecture and Design Thinking	
Instructions: 1)Draw neat diagrams wherever applicable 2) All questions are compulsory 3) Assume suitable data wherever necessary and justify the same		

Que. No	Question	Max. Mark
Q1	Solve any FOUR	20
i)	What kind of design decisions are required to be taken at the time of finalizing the software architecture?	5
ii)	What is Architectural Degradation? What are the types for the same?	5
iii)	Differentiate between Software Components and Software Connectors.	5
iv)	What are different analysis techniques with reference to software architecture?	5
v)	What are Design Directions with reference to developing a software system?	5
vi)	From designing a software point of view, what is modification? Explain types of modifications in brief.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	What are the important characteristics of a software model?	5
ii)	Differentiate between ambiguity, accuracy and precision with reference to software model giving example	5
	OR	
Q2 A	What do you understand by views and viewpoints, with reference to a software model? What are commonly used viewpoints?	10
Q 2 B	Solve any ONE	10
i)	With an example, explain each of 4 "C"s of architectural analysis goals.	10
ii)	What are dimensions of architectural consistencies?	10

Que. No.	Question	Max. Marks
Q3	Solve any TWO	20
i)	Explain the service(s), dimension(s), sub-dimension(s) and value(s) associated with Data Access connectors.	10
ii)	List the types of connectors, example of each and the services provided by each of them?	10
iii)	What are the benefits of a first class connector? With example explain explicit and implicit software connectors.	10

Que. No.	Question	Max. Marks
Q4	Solve any TWO	20
i)	With an example, explain in brief the seven stages of design process.	10
ii)	Giving examples, explain the types of Prototypes used in design	10
iii)	With reference to Thinking in signs, explain the terms Semiotics, Denotation and Cognition.	10

Que. No.	Question	Max. Marks
Q5	Attempt any FOUR	20
i)	What do you understand by Scope of Software Analysis?	5
ii)	What is the process of selection of a software connector?	5
iii)	With reference to developing designs, explain the terms adaptability, stories, and flexibility.	5
iv)	What are the key forms of appropriations in context of designing?	5
v)	What kind of attention is needed to be given before selecting colors in design?	5
vi)	What are the types of research carried out before designing a system?	5